Enrollment No.: _____

Darshan UNIVERSITY

Darshan Institute of Computer Application
BCA / B.Sc. (IT) | Sem-2 | Summer-2024

Course Code	:	2304CS205	**Date**	:	10-05-2024
Course Name	:	Operating System	**Duration**	:	150 Minutes
			Total Marks	:	70

Instructions:
1. Attempt all the questions.
2. Figures to the right indicates maximum marks.
3. Make suitable assumptions wherever necessary.

Q.1 **(A)** Define operating system. Write the goals of operating systems. **4**

(B) Discuss role of OS as a resource manager. **3**

OR

Compare multiprogramming systems and multiprocessing systems.

(C) List out different services provided by operating system and explain any four services in brief. **7**

OR

List out types of operating system and explain any two types in details.

Q.2 **(A)** Define process. Explain five state process state transition diagrams with the help of examples. **4**

(B) Compare Process and Thread. **3**

OR

Explain the process control block (PCB) with a diagram.

(C) Suppose that the following processes arrive at the times indicated. Each process will run for the listed amount of time. **7**

Process	Arrival Time (T0)	Time required for completion (ΔT) (CPU Burst Time)
P0	0	6
P1	1	3
P2	3	2

What is the average turnaround time and average waiting time using the following scheduling algorithm? (i) FCFS (ii) SJF

OR

Consider the following set of processes with the length of the CPU burst time given in seconds.

Process	Time required for completion (ΔT) (CPU Burst Time)
P0	6
P1	4
P2	2
P3	5

The processes are assumed to have arrived in the order p0, p1, p2, and p3 all at time 0. Draw gantt chart and calculate the average waiting time and average turnaround time using

1. round-robin scheduling (quantum=2)
2. round-robin scheduling (quantum=3)

Q.3 **(A)** List and explain conditions that lead to deadlock. **4**

(B) Define following terms: Race condition, Mutual exclusion, and Critical section. **3**

OR

Discuss the requirements for a solution to the critical section problem.

(C) Compare: Independent Process & Co-operating Processes. Write the advantages and Disadvantages of Inter Process Communication (IPC). **7**

OR

What is Deadlock? List out deadlock prevention techniques. Explain any two in details.

Q.4 **(A)** Explain memory partition: Fixed partition and Variable partition. **4**

(B) Compare Logical address and Physical address. **3**

OR

What is paging? Discuss basic paging techniques in detail.

(C) Given six memory partitions of 300 KB, 600 KB, 350 KB, 200 KB, 750 KB, and 125 KB (in order), how would the first-fit, best-fit, and worst-fit algorithms place processes of size 115 KB, 500 KB, 358 KB, 200KB, and 375 KB (in order)? Which algorithm is efficient for the use of memory? **7**

OR

Consider the following page reference string:
4, 7, 6, 1, 7, 6, 1, 2, 7, 2
How many page faults would occur for the following page replacement algorithm, considering 3 frames.
i) FIFO, and ii) Optimal

Q.5 **(A)** Write the attributes of file. **4**

(B) Define relative path and absolute path with an example. **3**

OR

Define directory. List out types of directories. Explain anyone in detail.

(C) What are the types of Files? Describe all the file operations. **7**

OR

Consider an imaginary disk with 51 cylinders. A request comes in to read a block on cylinder 10. While the seek to cylinder 10 is in progress, new requests come in for cylinders 2, 26, 16, 44, 9, and 12, in that order. Starting from the current head position, what is the total distance (in cylinders) that the disk arm moves to satisfy all the pending requests, for each of the following disk scheduling Algorithms

i. FCFS/FIFO
ii. SSTF

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